

# Engineering Membrane Platforms to Decode Receptor Nanodomains

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The plasma membrane is a highly organized and dynamic interface where proteins and lipids assemble into nanoscale functional domains that regulate signalling, trafficking, and cellular responses. Rather than being randomly distributed, membrane proteins form transient nanodomains whose molecular composition and spatial organization critically influence receptor activity. Although super-resolution microscopy has revolutionized the visualization of these structures, characterizing their composition across entire cell populations remains a major challenge, highlighting the need for complementary population-scale approaches.

My research focuses on developing integrated membrane-based platforms that combine DNA nanotechnology, biomimetic membrane systems, cell membrane, and quantitative biophysical methods to decipher the molecular organization of membrane receptors within their native nanoenvironment. To address the limitations of microscopy-only approaches, I developed NanoDeep<sup>1</sup>, a non-imaging strategy that translates membrane protein organization into a DNA sequencing readout through programmable DNA nanoassemblies. This platform enables systematic, population-scale mapping of the protein inventory surrounding selected membrane receptors while providing access to information on nanoscale organization. As a proof of concept, the platform is being applied to HER2, a clinically relevant receptor in breast cancer whose signalling is increasingly recognized to depend not only on expression levels but also on the composition and organization of its surrounding membrane nanoenvironment. Particular attention is devoted to specialized membrane domains such as lipid rafts, where receptor clustering and local lipid composition can critically influence signalling, drug sensitivity, and resistance mechanisms.

A central component of this research is the development of artificial membrane platforms based on supported lipid bilayers (SLBs) and pore spanning membranes (PSMs) that faithfully reproduce controlled membrane environments for mechanistic studies. These SLBs are generated through the fusion of Giant Unilamellar Vesicles (GUVs), for which we are developing a dedicated electroformation device based on gold interdigitated electrodes. Our current efforts are focused on optimizing these biomimetic membrane systems that can be integrated with DNA nanostructures to provide a robust framework for investigating how membrane composition and nanoscale organization influence receptor clustering and molecular interactions.

More broadly, this research seeks to establish a versatile toolkit for studying membrane receptor organization beyond Her2, providing new opportunities to identify molecular signatures that better define receptor functional states. By combining programmable DNA nanotechnology with engineered membrane systems and quantitative biophysics, the long-term goal is to uncover the membrane-level mechanisms that govern cellular signalling and therapeutic response across a broad range of biological contexts.

[1] Ambrosetti et al. 2021. DNA-nanoassembly-based approach to map membrane protein nanoenvironments. *Nat. Nanotechnol.* 16:85-95.